CSS 342: Data Structures, Algorithms, and Discrete Mathematics I

Overview

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- **Office hours:** Tue 3-4pm, Wed 4-5pm @ UW1-260Q - signup via Canvas Calendar
- **Class Sessions:** 1:15-3:15 MW
- **Prerequisite:** A minimum grade of 2.8 in either CSS 133, CSS 143, CSE 143, or CSS 162; and minimum grade of 2.5 in either STMATH 125 or MATH 125.
- Course website
- You can submit anonymous feedback for this class anytime during the quarter

Course Catalog Description: Integrating mathematical principles with detailed instruction in computer programming. Explores mathematical reasoning and discrete structures through object-oriented programming. Includes algorithm analysis, basic abstract data types, and data structures.

Goals

This fast-paced course is intended to enable students to design solutions to programming problems using object-oriented techniques. The course integrates the fundamental discrete mathematics aspects of computing with detailed instruction in end-to-end software design.

By the end of this quarter, you will be familiar with much of the C++ language and the basics of object-oriented programming. You will understand how to analyze a problem and design a solution. You will understand the basic data structures and algorithms which are the basis for much of computer science. You will also be able to analyze the trade-offs among memory, execution time, and implementation complexity associated with them. Topics include: recursion, computational complexity and algorithm analysis, logic, induction, lists, stacks, queues, sorting and searching. Also covered are object oriented fundamentals such as abstraction, encapsulation, hierarchy, and polymorphism.

Textbooks

- [Carrano] Frank M. Carrano and Timothy M Henry, Data Abstraction and Problem Solving with C++, Seventh edition, Addison-Wesley, 2017.

- [Cusack] Charles A. Cusack and David A. Santos, Active Introduction to Discrete Mathematics and Algorithms, Version 2.6.3, March 30, 2018. (online)

- [Chen] William Chen, Discrete Mathematics (online)

UW library also has a large collection of online books that are related, see Related Books & References as a starting point.

Details

Each lecture will be a combination of introducing new concepts and exercises to put these concepts into practice. We will complete lots of in-class Leetcode questions, so bring your computer to class and get ready to do some fast programming. Leetcode questions are like puzzles, so it is perfectly normal to feel confused and feel like you are not making any progress. You will get better over time. We will go over the solution to each problem, so make sure you ask questions if the answer is still not clear.

Take notes during class using pen and paper. After class, take time to go over your notes and reflect on the new concepts.

Reflecting on what you have learned is necessary to solidify what you have learned. Think about the following questions

- Why did we learn this concept?
- How does this concept relate to everything we have seen so far?
- Can I describe this concept in my own words?
- Write down some "tips" for if you were going to help a colleague who was struggling with this content.

Research shows that using pen and paper to take notes is much more effective than using a computer. Computers are often distracting to you and also to others around you. I recommend keeping a dedicated notebook for this course and using the existing techniques for reflecting on content learned to help organize the material and your thoughts.

Expect to spend 10-15 hours per week outside class on average.

Inclusion

All students are welcome in CSS 342 and are entitled to be treated respectfully by both classmates and the course staff. I strive to create a challenging but inclusive environment that is conducive to learning for all students. If at any time you feel that you are not experiencing an inclusive environment, or you are made to feel uncomfortable, disrespected, or excluded, please report the incident so that we may address the issue and maintain a supportive and inclusive learning environment. You may contact me or the CSS Advisors to express your concerns. Should you feel uncomfortable bringing up an issue with a staff member directly, you may also consider sending anonymous feedback

Required Course Work

There will be three categories of required course work:

- **Exercises (15%)**

Short assignments and activities some to be performed in-class and some outside of class. Exercises will be graded as Complete/Incomplete. If you are making a good faith effort and submitting the exercise by the due date, you will receive a grade of Complete. Most students who continue to be engaged in the class have an exercise grade of 90-100%. The exercises help prepare you for the exams. Your 2 lowest exercise grades will automatically be dropped in Canvas.

If you are not able to attend class for any reason (personal, professional, COVID quarantine, mental health day, etc), you can email me **before the start of class time** to be exempt from that day's in-class exercises.

You can complete up to 5 bonus projects to increase your exercise grade. The report for the bonus project must be submitted within 1-week of completing the activity.

- **Projects (35%)**

We will have 5 programming assignments. The programming assignments are sufficiently complex that you should not expect to complete them in one sitting. Start designing and thinking about the project as soon as it is assigned. Ask clarifying questions in class and on discord as necessary. Do not wait until the last day to start your project. Projects have very specific completion and submission requirements. You will need to give yourself enough time to package your project, check that you have completed all the components before submitting. You can submit your project multiple times on Canvas before the due date. I only look at the final submission. Most students have a project grade of 80-100%. The most common reason for low project grade is not starting the project on time. For one and only one project, you can have a 24-hour extension. To indicate you are using the extension, submit a brief explanation to the "Using Extension" project.

- **Exams (25% + 25%)**

We will have a midterm and a final exam to be held during the regular class session. The exams are closed-book, closed-notes exam, except for a single cheat sheet (single sheet of letter-sized, double-sided notes created by you). In an exam, you have to demonstrate your knowledge in limited time without any additional resources. The average for exams is usually 70-75%.

Late assignment will only be accepted when it is due to exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% will receive a minimum of 2.0, 85% will receive a minimum of 3.0 and 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Getting Help

There are multiple ways of getting help:

- **Discord**

With an active class, discord can be one of the easiest and fastest ways to get help. Discord channel is an extension of our class and the same rules for in-class behavior apply to discord. I will also monitor the discord channel and contribute to the discussion from time to time. When asking questions:

- **Do not** post code.
- Be clear in stating the problem. Explain how you tried to solve it. Make sure the answer is not the first link that comes up in a Google search.
- You can post a warning/error message from compiler and ask how others have resolved it.
- You can ask how a specific C++ function gets used

- You can discuss your approach in high-level terms

- If you post a question and then figure out the answer, post your answer so it benefits others in the class

In general, if you can discuss the problem in English without having to show code, you can discuss it on Discord.

- **QSC**

Quantitative Skills Center has dedicated hours with 342 tutors that can help. Here is a video about QSC services and the tutoring hours.

- **Office Hours**

If you need extra time after office hours and cannot get your issues resolved over discord, you can setup a meeting over zoom by sending me email (faiahmed@uw.edu). I will not debug your code, but I will answer questions and try to help you figure out the problem. I will always ask what you have already done so far to solve the problem, so come prepared.

If you have a personal matter, please see me in office hours.

- **Email**

Email is not an effective way to get help. Unless your question is of personal nature, it should be posted on discord. If you believe the question really needs my attention, you can tag me on discord.

Extenuating Circumstances

I recognize that our students come from varied backgrounds and can have widely-varying circumstances. If you have any unforeseen circumstances that arise during the course, please do not hesitate to contact me to discuss your situation. **The sooner I am made aware, the more easily I can provide accommodations.** While some amount of "productive struggle" is healthy for learning, you should ask me for help if you have been stuck on an issue for a very long time.

Life happens! While our focus is providing an excellent educational environment, our course does not exist in a vacuum. My ultimate goal is to provide you with the ability to be successful, and I encourage you to work with me to make that happen. Please don't suffer in silence.

Academic Integrity

Most exercises and projects in this course will be completed individually. Talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct: http://www.washington.edu/cssc/for-students/student-code-of-conduct/ Any code or text that you copy from an external source must have a comment (or reference) indicating its origins.

Access and Accommodations

Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 425.352.5307 or rosal@uw.edu.

DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

For Our Veterans

Welcome! We at UW Bothell understand that the transition into civilian life can be challenging for our veteran students and we have many resources for any who may want to reach out for guidance or assistance. This includes our Vet Corp Navigator through the WDVA and our Student Veterans Association (SVA). Please contact Veteran Services at 425.352.5307 or rosal@uw.edu. For those of you needing more URGENT support, please call The Suicide Prevention Hotline 1.800.273.8258 or connect with the UWB CARE Team https://www.uwb.edu/studentaffairs/care-team.

Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UWs policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form]().

Common Course Policies for the School of STEM

See https://bit.ly/STEMClassroom_Policies for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as ‘Professor Ahmed’)

Internships

Internships: One of the strengths of the computing industry is a culture that values internships. As a result, many industry leaders have well-developed internship programs for students to build their expertise and pathways to opportunities. Internships are offered throughout the year, some are intended for students just starting out (or just exploring) the technology industry, and many for the summer between your junior and senior year. **We strongly encourage you to participate in internships and attend campus Career Fairs and employer events to connect with industry partners and learn more about opportunities, hiring timelines and application processes.**

The common technical recruiting cycle is to hire in the fall for the next summer's internships and jobs. For some internships, this means the technical interview occurs 6-8 months before the actual internship. Thus, it is important to convey (in your application and interview) what your current knowledge base is at the time of the interview, and what it will be at the time of the internship. Because both of the Data Structures and Algorithms courses prepare you best for technical interviews, we recommend you do the following to be successful in the fall recruiting process:

- Check what the expectations are for the internship you are applying for. This includes both expectations at the time of the internship as well as at the time of the interview. The majority of internships are meant for juniors, but some are meant for graduating seniors.

- **Clearly communicate your qualifications,** being careful to accurately list your preparation and qualifications. It is important that you are able to back up the claims you make about your experience and skills from your application materials in behavioral and technical interviews. Misrepresentation of these reflects poorly on candidates and their future opportunities.

- We encourage you to take CSS 342 the fall quarter before the summer internship to prepare you better for the technical interviews.

- On your resume, list each course and the date you will be taking them. Thus, a student starting in Fall 2022 would list, for example: Data Structures, Algorithms, and Discrete Mathematics I (completing Fall 2022), Data Structures, Algorithms, and Discrete Mathematics II (completing Winter 2023).

- When you are invited for a technical interview, let the recruiter know where you are in the course series.

- To fill-in your content knowledge, prepare and practice for your technical interviews by utilizing the resources provided by Career Services at https://www.uwb.edu/career-services/interviews/technical-interviews.

There are many students who use internships for their Capstone course. However, this is **not** the only time when a student can do an internship. Rather, we encourage students to participate in **more than one experiential learning opportunity,** and internships can be done independent of receiving credit. If you decide to do a non-Capstone internship for credit, you can receive up to 5 cr of CSS 397, which will count as a 200-level CSS elective.

Finally, as a reminder, if you are considering an internship for your Capstone and you are a:

- CSSE major: You must first complete the Capstone prerequisites in order to use your internship for CSS 497. Prerequisites are completion of all core courses and 2 CSS electives. Most CSSE students will **not** have met the capstone prerequisites by their first summer. As mentioned earlier, these prerequisites are not a requirement for doing a non-Capstone internship (CSS 397) for credit.

- Applied Computing major: You cannot use an internship for your Capstone. You can, however, use an internship to earn up to 10 cr of CSS 495 which will count towards your 400 level CSS electives.

Your faculty, advisors, and Career Services are all great resources to help you be successful in your career pursuits.

- Make an appointment with your FYPP Academic Advisor (uwbadvis@uw.edu) or a CSS Academic Advisor (cssadv@uw.edu).

Make an appointment with Career Services at https://www.uwb.edu/career-services/appointments.